III B.TECH – I SEMESTER

Object Oriented Analysis And Design

Professional Elective-I | JNTUA R23 · B.Tech CSE

	OBJECT ORIENTED ANALYSIS AND DESIGN (Professional Elective –I)	L	T	P	C
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Course Objectives:

- Describe the activities in the different phases of the object-oriented development lifecycle.
- Understand the concepts of object-oriented model with the E-R and EER models.
- Model a real-world application by using UML diagram.
- Design architectural modelling.
- Describing an application of UML.

Course Outcomes:

At the end of the course, student will be able to

1. The importance of modelling in UML.

- Compare and contrast the object-oriented model with the E-R and EER models.
- Design use case diagram. Design an application using deployment diagram.
- Apply UML diagrams to build library application.

UNIT – I ** ** **9 Hrs**

Introduction to UML: Importance of modelling, principles of modelling, object-oriented modelling, conceptual model of the UML, Architecture, Software Development Life Cycle.

UNIT – II ** 9 Hrs **

Basic Structural Modelling: Classes, Relationships, common Mechanisms, and diagrams.

Advanced Structural Modelling: Advanced classes, advanced relationships, Interfaces, Types and Roles, Packages. Class & Object Diagrams: Terms, concepts, modelling techniques for Class & Object Diagrams.

UNIT – III ** ** **9 Hrs**

Basic Behavioural Modelling-I: Interactions, Interaction diagrams.

Basic Behavioural Modelling-II: Use cases, Use case Diagrams, Activity Diagrams.

UNIT – IV ** ** **9 Hrs**

Advanced Behavioral Modelling: Events and signals, state machines, processes and Threads, time and space, state chart diagrams. Architectural Modelling: Component, Deployment, Component diagrams and Deployment diagrams.

UNIT – V 9 Hrs

Patterns and Frameworks, Artifact Diagrams. Case Study: The Unified Library application.

Text Books:

- Grady Booch, James Rumbaugh, Ivar Jacobson: The Unified Modelling Language User Guide,

Pearson Education 2nd Edition.

- Object-Oriented Analysis and Design with the Unified Process By John W. Satzinger, Robert B

Jackson and Stephen D Burd, Cengage Learning.

Reference Books:

- Meilir Page-Jones: Fundamentals of Object-Oriented Design in UML, Pearson Education.
- Pascal Roques: Modelling Software Systems Using UML2, WILEY-Dreamtech India Pvt. Ltd.
- Atul Kahate: Object Oriented Analysis & Design, The McGraw-Hill Companies.
- Mark Priestley: Practical Object-Oriented Design with UML, TMH.
- Appling UML and Patterns: An introduction to Object – Oriented Analysis and Design and Unified Process, Craig Larman, Pearson Education.